

Telco Customer Churn Analysis

--All Customer Details

```
SELECT * FROM telco_customers;
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection
1	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No
2	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes
3	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No
4	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes
5	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No
6	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes
7	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No
8	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No
9	8580-PBAGW	Female	1	No	No	60	No	No	Fiber optic	No	No	No

--Total Customers

```
SELECT COUNT(*) FROM telco_customers;
```

	COUNT(*)
1	7043

--Unique Customers & Churned Customers

```
SELECT  
COUNT(DISTINCT customerID),  
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers  
FROM telco_customers;
```

	COUNT(DISTINCT customerID)	churned_customers
1	7043	1869

--Total Unique Customers

```
SELECT COUNT(DISTINCT customerID) FROM telco_customers;
```

	tenure_band	total_customers	churned_customers	churn_rate
1	0-1 Year	2069	999	48.28
2	1-2 Years	1141	332	29.1
3	2+ Years	3833	538	14.04

--Churned vs Non-Churned Customers

```
SELECT Churn, COUNT(*)  
FROM telco_customers  
GROUP BY Churn;
```

	Churn	COUNT(*)
1	No	5174
2	Yes	1869

--Churned Customers

```
SELECT COUNT(*) AS churned_customers  
FROM telco_customers  
WHERE Churn = 'Yes';
```

	churned_customers
1	1869

--Churn Rate %

```
SELECT  
ROUND(  
100.0*SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT(*),2 )AS churn_rate_percent  
FROM telco_customers;
```

	churn_rate_percent
1	26.54

--Active Customers

```
SELECT COUNT(*) AS active_customers
FROM telco_customers
WHERE Churn = 'No';
```

	active_customers
1	5174

--Avg Monthly Charges

```
SELECT ROUND(AVG(MonthlyCharges),2) AS avg_monthly_charges
FROM telco_customers;
```

	avg_monthly_charges
1	64.76

--Avg Tenure

```
SELECT ROUND(AVG(tenure), 1) AS avg_tenure_months
FROM telco_customers;
```

	avg_tenure_months
1	32.4

--Churn by Contract Type

```
SELECT * FROM churn_by_contract;
```

	Contract	total_customers	churned_customers	churn_rate
1	Month-to-month	3875	1655	42.71
2	One year	1473	166	11.27
3	Two year	1695	48	2.83

--Churn by Payment Method

```
SELECT
PaymentMethod,
COUNT(*) AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,
ROUND(
100.0 * SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT(*),2)AS churn_rate
FROM telco_customers
GROUP BY PaymentMethod;
```

	PaymentMethod	total_customers	churned_customers	churn_rate
1	Bank transfer (automatic)	1544	258	16.71
2	Credit card (automatic)	1522	232	15.24
3	Electronic check	2365	1071	45.29
4	Mailed check	1612	308	19.11

--Churn by Internet Service

```
SELECT
InternetService,
COUNT(*) AS total_customers,
SUM(CASE WHEN Churn = 'Yes' then 1 else 0 END) AS churned_customers,
ROUND(
100.0 * SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT(*),2
) AS churn_rate
FROM telco_customers
GROUP BY InternetService;
```

	InternetService	total_customers	churned_customers	churn_rate
1	DSL	2421	459	18.96
2	Fiber optic	3096	1297	41.89
3	No	1526	113	7.4

--Churn by Tenure Band

```
SELECT
CASE
WHEN tenure < 12 THEN '0-1 Year'
WHEN tenure <= 24 THEN '1-2 Years'
ELSE '2+ Years'
END AS tenure_band,
COUNT(*) AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,
ROUND(
100.0 * SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)/ COUNT(*),2) AS churn_rate
FROM telco_customers
GROUP BY tenure_band;
```

	tenure_band	total_customers	churned_customers	churn_rate
1	0-1 Year	2069	999	48.28
2	1-2 Years	1141	332	29.1
3	2+ Years	3833	538	14.04

--Top 10 Risk Customers

```
SELECT
customerID,
tenure,
MonthlyCharges
FROM telco_customers
WHERE Churn='Yes'
ORDER BY MonthlyCharges DESC
LIMIT 10;
```

	customerID	tenure	MonthlyCharges
1	8199-ZLLSA	67	118.35
2	2889-FPWRM	72	117.8
3	2302-ANTDP	48	117.45
4	9053-JZFKV	67	116.2
5	1444-VVSGW	70	115.65
6	0201-OAMXR	70	115.55
7	4361-BKAXE	41	114.5
8	1555-DJEQW	70	114.2
9	9158-VCTQB	41	113.6
10	7279-BUYWN	41	113.2